Assignment 7

Problems:

1. Let V be a finite-dimensional inner product space and T a linear operator on V. Show that the range of the adjoint T^* is the orthogonal complement of the null space of T.

2. If P is an orthogonal projection matrix, show that $I - P$ is a orthogonal projection matrix. Determine the range and nullspace of $I - P$.

3. Show that, if A has linearly independent columns, then

$$B = I - A\left(A^H A\right)^{-1} A^H$$

   is positive semidefinite, and hence that the minimum error $\mathbf{e}_{\mathrm{min}} = \mathbf{x} - P_A\mathbf{x} = B\mathbf{x}$ has an equal or smaller norm than the original vector $\mathbf{x}$. Hint: Consider $\|\mathbf{e}_{\mathrm{min}}\|^2 \geq 0$.

4. Formulate and solve the abstract least-squares regression problem

$$\min_{a_0,a_1,a_2} \sum_{i=1}^{n} |y_i - a_0 - a_1 x_i - a_2 x_i^2|^2$$

   in its linear form based on $\underline{e} = A\underline{c} - \underline{y}$. Use this to fit a parabola to the (x_i, y_i) data points $(-2,2)(-1,-10)(0,0)(1,2)(2,1)$. Make a plot showing the data on top of the fitted curve.

5. Define an inner product between matrices $X, Y \in \mathbb{C}^{l\times n}$ as

$$\langle X, Y\rangle = \operatorname{tr}\left(XY^H\right),$$

   where $\operatorname{tr}(\cdot)$ is the sum of the diagonal elements (see section C.3). We want to approximate the matrix Y by the scalar linear combination of matrices $X_1, X_2, \ldots, X_m$, as

$$Y = c_1X_1 + c_2X_2 + \cdots + c_mX_m + E.$$

   Using the orthogonality principle, determine a set of normal equations that can be used to find $c_1, c_2, \ldots, c_m$ that minimize the induced norm of E.

6. Using the dual approximation theorem, solve the finite dimensional problem

$$\begin{aligned} &\text{minimize } \mathbf{x}^H Q\mathbf{x} \\ &\text{subject to } A\mathbf{x} = \mathbf{b}, \end{aligned}$$

   where $\mathbf{x} \in \mathbb{C}^n$, Q is a positive-definite symmetric matrix, and A is an $m \times n$ matrix with $m < n$.